Sociotechnical Security

CS 8803/4803

Professor Veronica Rivera

Lecture 16 – Intimate Partner Violence II



Content warning: discussion of intimate partner violence (IPV), some high-level examples of vignettes centering on IPV, LGBTQ+ harms, and identity theft

Last time

- Attackers' view of IPV
  - Financial apps
  - Passkeys
- Introduced abusability framework
- Discussion of other areas where known-adversary threat model might apply

Today

- Responder's view of IPV
  - Clinic model
- Introduce trauma-informed computing
  - Approach used by Stephenson et al paper you all read
  - Uses tech-abuse clinic model as an example motivating the approach
- Work on mitigating researcher trauma

Abuse clinics as a data collection method

- Survivors' reactions to the tech abuse clinic
  - Diana Freed et al., (CSCW 2019)
- Usefulness of account security interfaces for identifying abuse
  - Alaa Daffalla et al., (USENIX Security 2023)

Abuse clinics are rare, can they scale?

- Abuse clinics are rare:
  - Technology-Enabled Coercive Control (TECC) clinic in Seattle, WA
  - Madison Tech Clinic in Madison, WI
  - Clinic to End Tech Abuse in NYC, NY
- Alternatives?
  - Could LLMs help?
  - In one study evaluating LLM responses for tech-facilitated abuse questions, 4 models tested produced ineffective, inaccurate, or incomplete responses in 81% of cases, and omitted critical safety warnings in 60% of responses
    - Vijay Prakash et al., Assessing LLM Response Quality in the Context of Technology-Facilitated Abuse, USENIX Security 2026

Trauma-informed computing

Digital technologies can create or exacerbate traumatic experiences

- Especially among certain groups: IPV survivors, sex workers, transgender/non-binary/LGBTQ+ people, racial minorities, people who have experienced the death of a loved one, etc.

Research questions

- How does trauma impact people's technology experiences?
  - How does tech facilitate trauma?
  - How does tech re-traumatize people?
- How should we design computing systems in light of trauma?

Janet X. Chen et al., Trauma-Informed Computing: Towards Safer Technology Experiences for All, CHI 2022

Vignette 1: IPV

- Jamie is an IPV survivor
- Her partner managed all of her tech and used that access to read her messages and monitor her activity.
- After she left, ordinary glitches like a fast-draining battery or an echo on calls make her fear he’s still watching.
- Her phone provider just tells her to reset everything, and she feels no one believes her.

Vignette 2: Gender transition

- Max sees online support during his gender transition
- Transphobic harassment on Reddit makes every notification a source of panic, so he deletes his account.
- Facebook recommends his second, private account to his family, outing him before he’s ready.
- Long after transitioning, emails using his deadname keep reopening the wound.

Vignette 3: Identity theft

- A scammer hijacks Alice’s phone number, takes over her accounts, and destroys her credit.
- Ashamed and blaming herself, she hides what happened from others.
- Years later, an Equifax breach notice throws her back into fear and hopelessness.

User behaviors through the lens of trauma

	Example Tech Experience	Possible Explanation Using the Lens of Trauma
Jamie	. . . assumes seemingly benign glitches are evidence of hacking.	Jamie may be **hypervigilant**, constantly scanning for signs of technology compromise as a defense.
	. . . is resigned to never having secure devices, accounts.	Jamie may be experiencing **anxiety** and **helplessness**.
Max	. . . stops using Reddit and turns off notifications.	Max may be **avoiding triggers** as a mechanism to prevent retraumatization.
	. . . angrily deletes Facebook account.	Max may be reacting to a traumatic event with **anger**, feelings of **hopelessness** about being safe online, and chooses to **withdraw**.
Alice	. . . is overwhelmed with fear by the breach notification.	Alice may be triggered by the new breach, leaving her feeling **fear**, **anxiety**, and **hopelessness**.
	. . . hides her experience from tech support.	Alice may be experiencing **shame** and **self-blame**.

Trauma-informed computing principles

Trauma-informed computing is an ongoing commitment to building and supporting technology that acknowledges trauma, recognizes tech can cause and worsen it, and works to avoid retraumatization.

Safety

People feel physically, digitally, and psychologically safe when using technology.

Trust

Systems are transparent, predictable, and reliable, and let users make mistakes and correct them.

Peer support

Help survivors connect with each other, with safeguards against abuse in those spaces.

Collaboration

Treat survivors as experts in their own lives, with a voice in support and in how tech is built.

Enablement

Give people the information and control to make their own informed choices.

Intersectionality

Account for the interlocking forms of oppression that shape people's experiences of trauma.

How to apply the TIC framework

Area	Topic	Example good practice
UX research & design	User research	Run studies somewhere participants already feel safe and familiar (safety, trust)
Security & privacy	Threat modeling	Treat “causing psychological distress” as a common adversarial goal (acknowledge trauma and its impact)
AI & machine learning	Intelligent agents	Make sure agents don’t judge users or heighten their anxiety (safety, trust)
Organizational culture	Work processes	Give workers training and resources to support survivors and to process secondary trauma (safety, peer support, enablement)

How the TIC framework has been applied

- Digital evidence collection for tech-facilitated abuse
  - Stephenson et al., (CHI 2026) **--> You read for class today**
- Social media design
  - Scott et al., (CHI 2023)
- Data donation platform for detecting online dating sexual violence
  - Zheng, Walquist, et al., (CHI 2024)
- Cybercrime recovery and support
  - Chen et al., (CHI 2026)

Mitigating trauma for researchers

Safer at-risk user research SoK

- Identifies risks of digital safety research to participants and researchers
- Existing practices
- Gaps & recommendations
  - Engage experts early
  - Use the lowest risk method that addresses the research goals

Applying TIC to qualitative analysis

- Motivation: qualitative research in digital safety risks traumatic exposure
- Question: How to enact TIC in software for mixed-initiative qualitative coding

Prototype: TIQA

- Design objective 1: Personalizable definitions of traumatic concepts
  - Analysts can flag concepts that are personally hard for them.
  - Import code embeddings from a trusted peer
- Design objective 2: Enable analysts to manage traumatic exposure
  - The tool tracks how much they've already been exposed to those concepts and for how long.
  - It estimates how much flagged content remains in the document
  - Ability for researchers to log that they are taking a break
    - Reaction prediction
    - Self-care nudges

Emily Tseng et al., Mitigating Trauma in Qualitative Research Infrastructure: Roles for Machine Assistance and Trauma-Informed Design, CSCW 2025

Logistics

- No mandatory midpoint check-in meeting with me
- Project presentations on Dec. 2 and Dec 7.
  - I will share a spreadsheet to sign up for a date by Friday on Ed
- New due date: Final project report will be due **Dec. 11 at 11:59pm**
- No class on Monday Nov. 23, Veronica will be away

Discussion

Warm-up: Identify one way the **Havron** paper's clinic model aligns with a trauma-informed computing principle, and one way the **Stephenson** paper applies a principle in designing Sherloc. For each, name the principle and explain the connection.